Lab 3: Component Modelling & Architectural Justification

Problem Statement #33: Micro-Lending & Peer Credit Risk Assessor

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Architecture Selection: "We chose a 3-Tier Layered Architecture with Service-Oriented Component Decoupling for the Micro-Lending & Peer Credit Risk Assessor System."

1. Architectural Choice

For the Micro-Lending platform, we selected a **3-Tier Layered Architecture** structured into: (1) Presentation & Client Layer, (2) Core Domain & Business Services Layer, and (3) Data Persistence & External Integration Layer. Internal communication between domain components (Loan Management, Credit Risk Engine, Amortization Engine, P2P Marketplace, Payment Settlement, and Default Monitor) is strictly governed by explicit provided and required interfaces using UML ball-and-socket notation (ICreditAssessment, IScheduleGenerator, IEscrowFunding, and IPaymentSettlement).

2. Two Specific Scenario-Related Reasons

• **Reason 1: Decoupling Volatile Heuristic Risk Logic from Core Settlement Ledgers (FR-001):**
Underwriting heuristic formulas (DTI weights, stability coefficients, credit tier thresholds) evolve continuously as risk policies are calibrated. Encapsulating the CreditRiskEngine behind the ICreditAssessment interface ensures algorithm tuning can occur without modifying database schemas, payment gateways, or double-entry accounting services.

• **Reason 2: Guaranteed Transactional Consistency in Fractional Escrow Payouts (FR-003, FR-004):**
In peer-to-peer lending, fractional lender allocations and installment dividend distributions must satisfy strict ledger conservation ($\sum$ Debits = $\sum$ Credits). A layered architecture enforces strict transactional boundaries, ensuring all balance mutations occur strictly within the Payment Settlement Service and the underlying ACID relational database.

3. Security Advantage: Sensitive Financial PII & Escrow Isolation

Borrower PAN, bank credentials, and credit reports represent high-risk data subject to RBI NBFC-P2P regulatory mandates. Layering ensures client-facing presentation apps have zero direct database access. All data queries must pass through authenticated business services where role-based access control (RBAC) and AES-256 column-level encryption are strictly enforced.

4. Performance Benefit: Caching Read-Heavy Marketplace Traffic Without Write Contention

Read-heavy operations (hundreds of peer lenders browsing loan listings) are separated from write-intensive ACID transaction ledgers. Marketplace listings are served from an in-memory Redis cache at the application layer, ensuring 99th-percentile repayment transaction latency remains strictly below 250 ms without lock contention on the primary relational database.